

THISEN LAKDINU

Department of Computer Engineering, University of Peradeniya, Sri Lanka

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ABOUT ME

Motivated Computer Engineering undergraduate with strong interests in embedded systems, networking, and software development. Experienced in academic and personal projects, event coordination, and leadership roles. Passionate about learning new technologies explore new uses cases of hardware and software and applying engineering knowledge to solve real-world problems.

EDUCATION

University of Peradeniya	Feb. 2023 – Present
<i>Bachelor of the Science of Engineering Honors (Computer Engineering) CGPA: 3.85/4.00</i>	Peradeniya, Sri Lanka
Relevant coursework: Programming Fundamentals (C, Java, Python), Data Structures and Algorithms, Object-Oriented Programming, Digital Logic Design, Computer Organization and Architecture, Embedded Systems and Microcontrollers, Computer Networks, Database Management Systems, Operating Systems, Software Engineering, Signals and Systems.	
D.S. Senanayaka College Colombo 07	Feb. 2008 – Oct. 2022(23)
<i>G.C.E. Advanced Level (Physical Science Stream) Results: 3 A's District Rank: 244 (Colombo)</i>	Colombo, Sri Lanka

RESEARCH EXPERIENCE

Autonomous Robotic Platform for Real-Time Plant Phenotyping	Jan 2026 – Present
<i>Faculty of Agriculture & Faculty of Engineering, University of Peradeniya, Sri Lanka</i>	
Computer Vision Developer - Phenotypic Monitoring Initiative	
- Engineered a native Android application for the non-destructive monitoring and estimation of plant phenotypic traits (Leaf Area), replacing traditional destructive sampling methods. - Implemented native C++ OpenCV algorithms via Java Native Interface (JNI) to process live 30fps camera feeds, utilizing ArUco markers for real-time perspective transformation and area calculation. - Developed local on-device data pipelines using SQLite and Android's Storage Access Framework to export continuous field data (CSV) for agricultural research. - Resolved complex memory management constraints in mobile environments by writing dynamic matrix-release protocols for heavy image-processing loops.	

TECHNICAL PROJECT EXPERIENCE

Modular Aquarium Management System

2026 – Present

ESP32, IoT, Sensors, MQTT, Web Dashboard, Js

Designed and implemented a modular IoT-based aquarium automation and monitoring system enabling real-time tracking of temperature, water level, pH, and turbidity.

- Key Contributions:
 - Developed ESP32 firmware for multi-sensor integration and reliable data acquisition.
 - Implemented MQTT-based communication architecture for real-time data transmission across distributed network nodes.
 - Designed backend data pipeline and web-based dashboard for live visualization and alert-based monitoring.
 - Engineered modular hardware design allowing scalable sensor expansion and easy maintenance.

Courier Delivery Tracking and Management System

2025

MySQL, SQL, Database Design

Designed and implemented a complete relational database system for managing courier operations including parcels, vehicles, employees, routes, and customers.

- Key Contributions:
 - Developed fully normalized schema with integrity constraints and optimized indexing strategies.
 - Populated database with 20+ meaningful records for realistic system testing and performance validation.
 - Implemented advanced SQL queries, views, triggers, and stored procedures for automation and reporting.

Traffic Light Controller with Pedestrian Signals

2025

C, Embedded Systems, FSM Design

Designed and implemented a finite-state-machine-based traffic light control system with integrated pedestrian crossing signals.

- Key Contributions:
 - Implemented timing-accurate phase transitions using structured C programming.
 - Added pedestrian interrupt handling and safe crossing logic for real-world traffic scenarios.
 - Validated design using simulation and hardware testing methodologies.

Hospital CKD Patient Data Entry & Management System

2025

Database Design, UI Workflow, System Architecture

Designed a scalable data entry and management platform to streamline Chronic Kidney Disease (CKD) patient record handling in hospitals.

- Key Contributions:
 - Designed relational database schema ensuring secure storage, consistency, and efficient retrieval of patient medical records.
 - Developed structured data entry workflows minimizing human error and improving usability for healthcare staff.
 - Focused on system scalability, privacy, and maintainability.

Volunteer Developer — University Development Team | University of Peradeniya 2025 – Present

Web Development, Website Maintenance, UI Improvements, Content Management

Contributed as a volunteer developer in maintaining and improving departmental websites and web pages to enhance accessibility, usability, and overall user experience.

Key Contributions:

- Maintained and updated departmental websites with new content, announcements, and structural improvements.
- Assisted in improving website responsiveness, usability, and page performance.
- Collaborated with development team members to troubleshoot issues and implement website enhancements.
- Supported continuous maintenance and optimization of university department web platforms.

Peraverse - Full-Stack Integration Platform for crowd management 2026 – Present

React, TypeScript, Node.js, Express.js, PostgreSQL, Supabase, Docker, REST APIs, Git, Project Management

Designed and developed a scalable real-time spatial analytics and content management platform for the PeraMorphiq neuromorphic computing group at the University of Peradeniya, focusing on system integration, performance, and maintainability.

Key Contributions:

- Refactored and integrated legacy architecture by consolidating seven disparate microservices into a single unified Express.js backend, reducing operational overhead and standardizing API structures across the system.
- Containerized the full platform using Docker and Docker Compose, enabling consistent multi-container development environments and streamlining deployment workflows with full local-to-cloud parity.
- Coordinated end-to-end development workflows by aligning frontend (React/TypeScript) and backend deliverables, managing task allocation, and driving sprint execution to meet architectural milestones.
- Enforced Git workflow standards, including branching strategies, code reviews, and quality control gates, improving team collaboration and code reliability.
- Developed a resilient bidirectional data synchronization engine between PostgreSQL and Supabase, implementing automated recovery mechanisms and real-time health monitoring endpoints (`/api/heatmap/sync-status`).
- Resolved critical production and scalability issues including memory leaks, CORS failures, database connection drops, and data synchronization loop bugs under high-load simulation scenarios.

PROFESSIONAL EXPERIENCE

Undergraduate Teaching Assistant (Part-time)

Feb 2024 – Present

Department of Computer Engineering, University of Peradeniya

- Conducting and guiding laboratory sessions, for Computer Architecture (CO2070), Neetwork design principles (CO2080), Digital Design (CO2010), Computer Networks (CO2030 & CO323), and Computing (Python) (CO1010, GP106) for junior undergraduate batches.

Secretary

2025 – Present

Batch representative

2023 – 2025

IESL Students' Chapter, University of Peradeniya

- Managing official documentation, correspondence, meeting records, and administrative operations of the student chapter.
- Coordinating planning and execution of technical workshops, seminars, competitions, industrial visits, and professional development programs.
- Acting as the middle between academic staff, industry professionals, and student members to ensure smooth event execution.

Assistant Secretary (2024 - 2025)

Association of Computer Engineering Students (ACES)

Committee Member (Batch Rep) (2024 - 2025)

Association of Computer Engineering Students (ACES)

Assistant Treasury (2025 - Present)

Director Board member (2024-2025)

Leo Society, University of Peradeniya

Committee Member (2024 - 2025)

Junior Treasury (2026- Present)

Buddhist Society, University of Peradeniya

Project Manager (2026 - Present)

IEEE Computer science Society, University of Peradeniya

TECHNICAL SKILLS

Languages: Python, Java, C/C++, Verilog HDL

Frameworks & Dev: Flutter, Spring Boot, Android Studio, Expo

Databases: MongoDB, MySQL, Postgres, Firebase, Supabase

ML & Computer Vision: PyTorch, OpenCV, CNN, Lama

Embedded & Signal: ESP32, MATLAB Signal Analysis, Postgres

Tools: Git, LaTeX, Jupyter, Google Colab, Postman, Gitlab

REFERENCES

Prof. Roshan Ragel

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